Integrated Textbook Notes for Advanced Engineering Mathematics

Based on LN01–LN09 Lecture Notes

Compiled Study Notes

Contents

1 Vectors in Two- and Three-Dimensional Space **1**
1.1 Geometric Vectors . . . 1
1.2 Vector Addition and Subtraction . . . 1
1.3 Vectors in a Two-Dimensional Coordinate System . . . 2
1.4 Vectors in Three-Dimensional Space . . . 3
1.5 Dot Product . . . 3

2 Cross Products, Lines, Planes, and Bases **5**
2.1 The Cross Product . . . 5
2.2 Scalar Triple Product and Vector Triple Product . . . 7
2.3 Lines in Space . . . 8
2.4 Planes . . . 8
2.5 Bases and the Gram–Schmidt Process . . . 9

3 Matrix Algebra and Linear Systems **11**
3.1 Matrices and Matrix Operations . . . 11
3.2 Special Matrices . . . 12
3.3 Systems of Linear Algebraic Equations . . . 12
3.4 Rank of a Matrix . . . 13

4 Determinants and Eigenvalues **15**
4.1 Determinants . . . 15
4.2 Inverse Matrices . . . 18
4.3 Cramer's Rule . . . 19
4.4 Eigenvalue Problems . . . 20
4.5 Powers of Matrices and the Cayley–Hamilton Theorem . . . 22
4.6 Diagonalizability . . . 23
4.7 Orthogonal Matrices and Orthogonal Diagonalization . . . 24
4.8 LU-Factorization . . . 25

5 Laplace Transforms **27**
5.1 Definition and Linearity . . . 27
5.2 Basic Transform Table . . . 27
5.3 Transforms of Derivatives . . . 28
5.4 Solving Initial-Value Problems . . . 28
5.5 Shift on the s-Axis . . . 28
5.6 Unit Step Function . . . 28

6 Vector Functions, Motion, and Partial Derivatives **31**
6.1 Vector Functions . . . 31
6.2 Length of a Curve . . . 32
6.3 Motion on a Curve . . . 32

6.4 Partial Derivatives . . . 33
6.5 Chain Rule for Several Variables . . . 33

7 Line Integrals and Green's Theorem **35**
7.1 Line Integrals . . . 35
7.2 Conservative Fields . . . 36
7.3 Double Integrals . . . 36
7.4 Polar Coordinates . . . 36
7.5 Green's Theorem . . . 37

8 Complex Numbers and Complex Functions **39**
8.1 Complex Numbers . . . 39
8.2 Powers and Roots . . . 39
8.3 Functions of a Complex Variable . . . 40
8.4 Limits, Continuity, and Derivatives . . . 40
8.5 Cauchy–Riemann Equations . . . 41
8.6 Integration in the Complex Plane . . . 41

9 Additional Topics **43**
9.1 Additional Determinant Properties . . . 43
9.2 Additional Laplace Transform Operations . . . 43
9.3 Directional Derivatives and Gradients . . . 44
9.4 Level Curves, Level Surfaces, Tangent Planes, and Normal Lines . . . 44

Formula Summary **45**

List of Figures

1.1 Vector addition can be visualized either by placing vectors head-to-tail or by completing a parallelogram. 2
1.2 The dot product determines the angle between two vectors and the projection of one vector onto another. 4

2.1 The vectors $\mathbf{a}$ and $\mathbf{b}$ span a parallelogram. Its area is $\|\mathbf{a} \times \mathbf{b}\|$, and the vector $\mathbf{a} \times \mathbf{b}$ is perpendicular to the plane spanned by $\mathbf{a}$ and $\mathbf{b}$. The coordinate axes are schematic. 6
2.2 A line is determined by one point and one direction vector. 8
2.3 A plane is described by a point on the plane together with a normal vector. . . . 9
2.4 Gram–Schmidt subtracts the projection of $\mathbf{v}_2$ onto $\mathbf{u}_1$ to produce an orthogonal vector $\mathbf{u}_2$. 10

4.1 A 2×2 matrix A transforms the unit square into the parallelogram spanned by its column vectors $A\mathbf{e}_1$ and $A\mathbf{e}_2$. Since the unit square has area 1, the area of the transformed parallelogram is $|\det A|$. Thus $|\det A|$ measures the area scaling factor of A. 18
4.2 An eigenvector is a direction preserved by a linear transformation. The vectors $\mathbf{v}$ and $A\mathbf{v}$ lie on the same line, so $\mathbf{v}$ is an eigenvector and $A\mathbf{v} = \lambda\mathbf{v}$ for some scalar λ. By contrast, $\mathbf{w}$ and $A\mathbf{w}$ do not have the same direction, so $\mathbf{w}$ is not an eigenvector. 22

5.1 The unit step function $U(t-a)$ switches from 0 to 1 at $t = a$. 28

6.1 A vector-valued function traces out a curve, and the derivative vector at a point gives the tangent direction. 31

8.1 Roots of unity in the complex plane. 40

9.1 The gradient is perpendicular to a level curve and points in the direction of greatest increase. 44

Preface

These notes reorganize the material from the uploaded lecture notes into a connected textbook-style document. The order is pedagogical rather than slide-by-slide: vectors and analytic geometry first, then matrices and linear systems, then transforms, vector functions and multivariable calculus, line and double integrals, and finally complex numbers and complex integration. Every major definition, formula, property, method, and example appearing in the lecture slides is incorporated in expanded prose. This revised version also fixes compilation issues, removes corrupted control characters, and expands several geometric explanations such as orientation, the right-hand rule, and domain assumptions for vector fields. In this final revision, all figures are redrawn with consistent label placement, white-backed annotations, and textbook-style captions for improved readability.

Chapter 1

Vectors in Two- and Three-Dimensional Space

Chapter Roadmap

This chapter develops vectors from three complementary viewpoints: geometry, coordinates, and algebra. The goal is to move fluently between arrows in space, component formulas, and geometric interpretations such as length, angle, projection, and orthogonality.

1.1 Geometric Vectors

Definition

A *vector* is a quantity having both magnitude and direction. If a vector has initial point A and terminal point B, it is denoted by $\overrightarrow{AB}$. Its magnitude is denoted by $\|\overrightarrow{AB}\|$.

Two vectors are equal if they have the same magnitude and the same direction. Vectors are *free*: a vector may be translated from one position to another as long as its magnitude and direction do not change. The negative vector $-\mathbf{a}$ has the same magnitude as $\mathbf{a}$ but the opposite direction.

If $k \neq 0$ is a scalar and $\mathbf{a}$ is a vector, then $k\mathbf{a}$ is the scalar multiple of $\mathbf{a}$. Its length is $|k|\|\mathbf{a}\|$, and its direction is the same as $\mathbf{a}$ if $k > 0$ and opposite if $k < 0$. Two nonzero vectors are parallel precisely when one is a nonzero scalar multiple of the other.

1.2 Vector Addition and Subtraction

The sum $\mathbf{a} + \mathbf{b}$ can be constructed by the triangle rule or the parallelogram rule. In the triangle rule, place the initial point of $\mathbf{b}$ at the terminal point of $\mathbf{a}$; the resulting vector from the initial point of $\mathbf{a}$ to the terminal point of $\mathbf{b}$ is $\mathbf{a} + \mathbf{b}$. Subtraction is addition of the negative vector:

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}).$$

From a pedagogical viewpoint, vector addition expresses the combined effect of two displacements. The triangle rule is convenient when one motion follows another, while the parallelogram rule is convenient when two vectors act from the same point. In engineering and physics, this idea appears constantly: forces combine by vector addition, velocity components combine by vector addition, and position changes accumulate by vector addition.

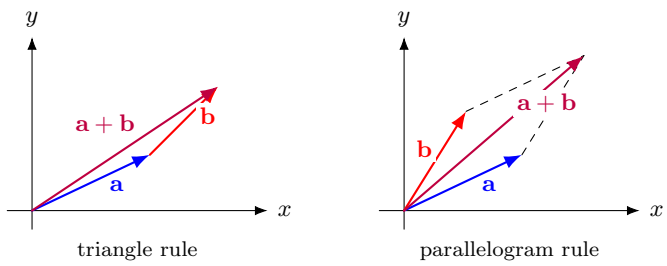

Figure 1.1: Vector addition can be visualized either by placing vectors head-to-tail or by completing a parallelogram.

1.3 Vectors in a Two-Dimensional Coordinate System

A point $P(x_1, y_1)$ determines the position vector

$$\overrightarrow{OP} = \langle x_1, y_1 \rangle.$$

A general vector in the plane is an ordered pair

$$\mathbf{a} = \langle a_1, a_2 \rangle,$$

where a_1 and a_2 are the components of the vector. For vectors $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$,

$$\mathbf{a} + \mathbf{b} = \langle a_1 + b_1, a_2 + b_2 \rangle, \quad \mathbf{a} - \mathbf{b} = \langle a_1 - b_1, a_2 - b_2 \rangle, \quad k\mathbf{a} = \langle ka_1, ka_2 \rangle.$$

The zero vector is $\mathbf{0} = \langle 0, 0 \rangle$.

Theorem / Property

For plane vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and scalars k, k_1, k_2,

$$\begin{aligned} \mathbf{a} + \mathbf{b} &= \mathbf{b} + \mathbf{a}, & \mathbf{a} + (\mathbf{b} + \mathbf{c}) &= (\mathbf{a} + \mathbf{b}) + \mathbf{c}, \\ \mathbf{a} + \mathbf{0} &= \mathbf{a}, & \mathbf{a} + (-\mathbf{a}) &= \mathbf{0}, \\ k(\mathbf{a} + \mathbf{b}) &= k\mathbf{a} + k\mathbf{b}, & (k_1 + k_2)\mathbf{a} &= k_1\mathbf{a} + k_2\mathbf{a}, \\ (k_1 k_2)\mathbf{a} &= k_1(k_2\mathbf{a}). \end{aligned}$$

The magnitude, length, or norm of $\mathbf{a} = \langle a_1, a_2 \rangle$ is

$$\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2}.$$

A vector of magnitude 1 is a unit vector. If $\mathbf{a} \neq \mathbf{0}$, then the unit vector in the direction of $\mathbf{a}$ is

$$\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|}.$$

The standard unit vectors in $\mathbb{R}^2$ are

$$\mathbf{i} = \langle 1, 0 \rangle, \qquad \mathbf{j} = \langle 0, 1 \rangle,$$

and every vector has the linear combination form

$$\mathbf{a} = \langle a_1, a_2 \rangle = a_1\mathbf{i} + a_2\mathbf{j}.$$

1.4 Vectors in Three-Dimensional Space

A rectangular coordinate system in $\mathbb{R}^3$ has mutually orthogonal x-, y-, and z-axes. These axes are oriented according to the right-hand rule. A point is represented by an ordered triple $P(a, b, c)$.

For $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$, the distance is

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

The midpoint of the segment joining P_1 and P_2 is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right).$$

A vector in $\mathbb{R}^3$ is

$$\mathbf{a} = \langle a_1, a_2, a_3 \rangle.$$

The standard unit vectors are

$$\mathbf{i} = \langle 1, 0, 0 \rangle, \qquad \mathbf{j} = \langle 0, 1, 0 \rangle, \qquad \mathbf{k} = \langle 0, 0, 1 \rangle,$$

and every $\mathbf{a} \in \mathbb{R}^3$ can be written as

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}.$$

The magnitude is

$$\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2},$$

and the unit vector in the direction of a nonzero $\mathbf{a}$ is $\mathbf{a}/\|\mathbf{a}\|$.

1.5 Dot Product

For $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$ in $\mathbb{R}^2$,

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2.$$

For $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ in $\mathbb{R}^3$,

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

Theorem / Property

The dot product satisfies

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \mathbf{b} \cdot \mathbf{a}, \\ \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) &= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}, \\ k(\mathbf{a} \cdot \mathbf{b}) &= (k\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (k\mathbf{b}), \\ \mathbf{a} \cdot \mathbf{a} &= \|\mathbf{a}\|^2. \end{aligned}$$

The geometric form of the dot product is

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta, \qquad 0 \le \theta \le \pi.$$

Therefore two nonzero vectors are orthogonal if and only if

$$\mathbf{a} \cdot \mathbf{b} = 0.$$

The scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ is

$$\operatorname{comp}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|},$$

and the vector projection is

$$\operatorname{proj}_{\mathbf{b}} \mathbf{a} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}} \right) \mathbf{b}.$$

The dot product measures directional agreement. If the angle θ is acute, then $\cos\theta > 0$ and the dot product is positive; if θ is obtuse, then the dot product is negative; and if $\theta = \pi/2$, then the dot product is zero. This interpretation is fundamental in projection, work, Fourier-type expansions, and orthogonality arguments.

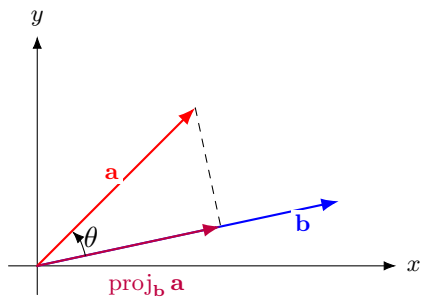

Figure 1.2: The dot product determines the angle between two vectors and the projection of one vector onto another.

Key Takeaways

- Vector addition and scalar multiplication are defined componentwise, but their geometric meaning is displacement and scaling.
- The dot product detects angle, projection, and orthogonality.
- Unit vectors separate direction from magnitude.

Chapter 2

Cross Products, Lines, Planes, and Bases

Chapter Roadmap

This chapter focuses on three-dimensional geometry. The cross product produces perpendicular directions and areas, while lines and planes are described using points, direction vectors, and normal vectors. The chapter ends with bases and the Gram–Schmidt process, which turn arbitrary independent vectors into orthogonal or orthonormal systems.

2.1 The Cross Product

For $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, the cross product is

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}.$$

Equivalently,

$$\mathbf{a} \times \mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\| \sin\theta\, \mathbf{n},$$

where $\mathbf{n}$ is the unit vector perpendicular to the plane of $\mathbf{a}$ and $\mathbf{b}$ whose direction is determined by the right-hand rule. Its magnitude is

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\| \sin\theta.$$

The right-hand rule, carefully explained

The cross product is not merely a formula; it encodes an orientation. To determine the direction of $\mathbf{a} \times \mathbf{b}$, place the tails of $\mathbf{a}$ and $\mathbf{b}$ at the same point and use the following rule:

1. Point the fingers of your right hand in the direction of the first vector $\mathbf{a}$.
2. Curl your fingers through the smaller angle from $\mathbf{a}$ toward the second vector $\mathbf{b}$.
3. Your thumb points in the direction of $\mathbf{a} \times \mathbf{b}$.

This is why order matters. Reversing the order reverses the orientation:

$$\mathbf{b} \times \mathbf{a} = -(\mathbf{a} \times \mathbf{b}).$$

In the standard right-handed coordinate system,

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \qquad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \qquad \mathbf{k} \times \mathbf{i} = \mathbf{j},$$

while reversing any pair changes the sign:

$$\mathbf{j} \times \mathbf{i} = -\mathbf{k}, \qquad \mathbf{k} \times \mathbf{j} = -\mathbf{i}, \qquad \mathbf{i} \times \mathbf{k} = -\mathbf{j}.$$

Geometrically, the magnitude $\|\mathbf{a} \times \mathbf{b}\|$ is the area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$. Therefore the cross product is zero exactly when the parallelogram has zero area, which happens when the vectors are parallel or one vector is zero.

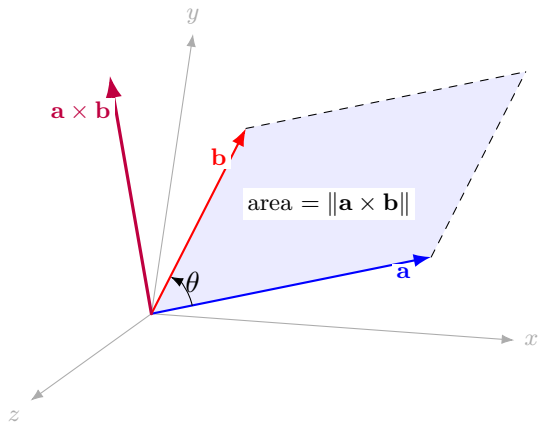

Figure 2.1: The vectors $\mathbf{a}$ and $\mathbf{b}$ span a parallelogram. Its area is $\|\mathbf{a} \times \mathbf{b}\|$, and the vector $\mathbf{a} \times \mathbf{b}$ is perpendicular to the plane spanned by $\mathbf{a}$ and $\mathbf{b}$. The coordinate axes are schematic.

Example

Let $\mathbf{a} = 4\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} + \mathbf{j} - \mathbf{k}$. Then

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -2 & 5 \\ 3 & 1 & -1 \end{vmatrix} = -3\mathbf{i} + 19\mathbf{j} + 10\mathbf{k}.$$

Theorem / Property

Properties of the cross product:

(i) $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ if $\mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$.

(ii) $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$.

(iii) $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.

(iv) $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$.

(v) $\mathbf{a} \times (k\mathbf{b}) = (k\mathbf{a}) \times \mathbf{b} = k(\mathbf{a} \times \mathbf{b})$.

(vi) $\mathbf{a} \times \mathbf{a} = \mathbf{0}$.

(vii) $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0$ and $\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 0$.

Two nonzero vectors are parallel if and only if $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.

How to choose dot product, cross product, or scalar triple product

Use the dot product when the question involves angles, orthogonality, projection, work, or a component in a direction. Use the cross product when the question involves a perpendicular vector, an area in three-dimensional space, or a normal vector to a plane. Use the scalar triple product when the question involves volume or checking whether three vectors lie in one plane.

2.2 Scalar Triple Product and Vector Triple Product

The scalar triple product is

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

The vector triple product identity is

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$$

The area of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$ is

$$A = \|\mathbf{a} \times \mathbf{b}\|,$$

and the area of the triangle with adjacent side vectors $\mathbf{a}$ and $\mathbf{b}$ is

$$A = \frac{1}{2}\|\mathbf{a} \times \mathbf{b}\|.$$

The volume of the parallelepiped determined by $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|.$$

Thus $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are coplanar if and only if

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0.$$

Example

For points $P_1(1, 1, 1), P_2(2, 3, 4), P_3(3, 0, -1)$, use side vectors

$$\overrightarrow{P_2P_1} = -\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}, \qquad \overrightarrow{P_2P_3} = \mathbf{i} - 3\mathbf{j} - 5\mathbf{k}.$$

Then

$$\overrightarrow{P_2P_1} \times \overrightarrow{P_2P_3} = \mathbf{i} - 8\mathbf{j} + 5\mathbf{k},$$

so the triangle area is

$$A = \frac{1}{2}\sqrt{1^2 + (-8)^2 + 5^2} = \frac{3\sqrt{10}}{2}.$$

Example

To determine whether four points $P_1(1, 1, -2)$, $P_2(4, 0, -3)$, $P_3(1, -5, 10)$, and $P_4(-7, 2, 4)$ lie in one plane, form

$$\overrightarrow{P_1P_2} = 3\mathbf{i} - \mathbf{j} - \mathbf{k}, \quad \overrightarrow{P_1P_3} = -6\mathbf{j} + 12\mathbf{k}, \quad \overrightarrow{P_1P_4} = -8\mathbf{i} + \mathbf{j} + 6\mathbf{k}.$$

If
$$\overrightarrow{P_1P_2} \cdot (\overrightarrow{P_1P_3} \times \overrightarrow{P_1P_4}) = 0,$$
then the points are coplanar. In the lecture example this value is 0, so the four points lie in the same plane.

2.3 Lines in Space

A line through point $P_0(x_0, y_0, z_0)$ with direction vector $\mathbf{a} = \langle a, b, c\rangle$ has vector equation
$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{a},$$
parametric equations
$$x = x_0 + at, \qquad y = y_0 + bt, \qquad z = z_0 + ct,$$
and, when $abc \neq 0$, symmetric equations
$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Example

The line through $(2, -1, 8)$ and $(5, 6, -3)$ has direction vector
$$\mathbf{a} = (2, -1, 8) - (5, 6, -3) = \langle -3, -7, 11\rangle.$$
A vector equation is
$$\mathbf{r} = 2\mathbf{i} - \mathbf{j} + 8\mathbf{k} + t(-3\mathbf{i} - 7\mathbf{j} + 11\mathbf{k}),$$
so
$$x = 2 - 3t, \qquad y = -1 - 7t, \qquad z = 8 + 11t,$$
and
$$\frac{x - 2}{-3} = \frac{y + 1}{-7} = \frac{z - 8}{11}.$$

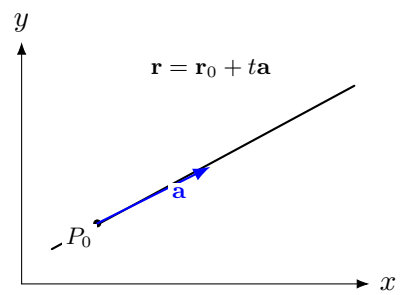

Figure 2.2: A line is determined by one point and one direction vector.

2.4 Planes

A plane containing point $P_1(x_1, y_1, z_1)$ and perpendicular to normal vector $\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ is described by
$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_1) = 0.$$

Its Cartesian equation is

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0.$$

To find a plane through three noncollinear points, construct two direction vectors in the plane and take their cross product as a normal vector.

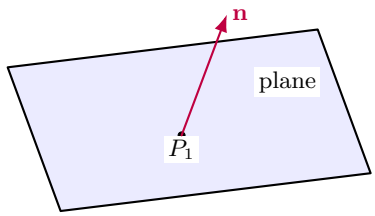

Figure 2.3: A plane is described by a point on the plane together with a normal vector.

2.5 Bases and the Gram–Schmidt Process

A set of linearly independent vectors that spans a vector space is a basis. In $\mathbb{R}^n$, every vector can be written as a linear combination of n linearly independent basis vectors. The standard basis is

$$\mathbf{e}_1 = (1, 0, 0, \ldots, 0), \quad \mathbf{e}_2 = (0, 1, 0, \ldots, 0), \quad \ldots, \quad \mathbf{e}_n = (0, 0, 0, \ldots, 1).$$

This is an orthonormal basis because

$$\mathbf{e}_i \cdot \mathbf{e}_j = 0 \quad (i \neq j), \qquad \|\mathbf{e}_i\| = 1.$$

Gram–Schmidt Orthogonalization

Given a basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$, Gram–Schmidt produces an orthogonal basis $B' = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ and then an orthonormal basis $B'' = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ by normalizing.
In two dimensions:

$$\mathbf{u}_1 = \mathbf{v}_1, \qquad \mathbf{u}_2 = \mathbf{v}_2 - \operatorname{proj}_{\mathbf{u}_1} \mathbf{v}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1.$$

In three dimensions:

$$\begin{aligned} \mathbf{u}_1 &= \mathbf{v}_1, \\ \mathbf{u}_2 &= \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1, \\ \mathbf{u}_3 &= \mathbf{v}_3 - \frac{\mathbf{v}_3 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 - \frac{\mathbf{v}_3 \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}\mathbf{u}_2. \end{aligned}$$

Finally,

$$\mathbf{w}_i = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}.$$

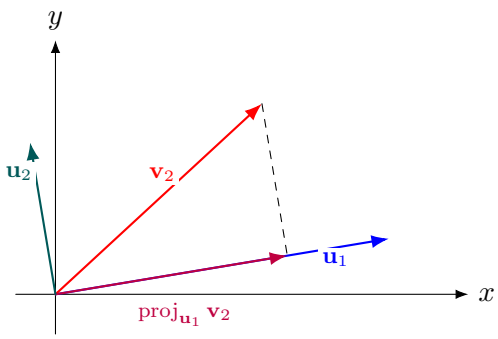

Figure 2.4: Gram–Schmidt subtracts the projection of $\mathbf{v}_2$ onto $\mathbf{u}_1$ to produce an orthogonal vector $\mathbf{u}_2$.

Key Takeaways

- Cross products produce normal vectors and areas; scalar triple products produce volumes and coplanarity tests.
- A line needs a point and a direction vector; a plane needs a point and a normal vector.
- Gram–Schmidt converts an arbitrary basis into an orthogonal or orthonormal basis by subtracting projections.

Chapter 3

Matrix Algebra and Linear Systems

Chapter Roadmap

Matrices organize linear systems and linear transformations. This chapter emphasizes how matrix operations are defined, how row operations preserve solution sets, and how rank determines consistency and the number of free parameters.

3.1 Matrices and Matrix Operations

Definition

A matrix is a rectangular array of numbers or functions. An $m \times n$ matrix has m rows and n columns. An $n \times n$ matrix is a square matrix of order n. A column vector is an $n \times 1$ matrix, and a row vector is a $1 \times n$ matrix.

Write

$$A = (a_{ij})_{m\times n} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

where a_{ij} is the entry in row i and column j.

For matrices of the same size,

$$A + B = (a_{ij} + b_{ij})_{m\times n}, \qquad kA = (ka_{ij})_{m\times n}.$$

Theorem / Property

For $m \times n$ matrices A, B, C and scalars k_1, k_2,

$$\begin{aligned} A + B &= B + A, & A + (B + C) &= (A + B) + C, \\ (k_1k_2)A &= k_1(k_2A), & 1A &= A, \\ k_1(A + B) &= k_1A + k_1B, & (k_1 + k_2)A &= k_1A + k_2A. \end{aligned}$$

If A is $m \times p$ and B is $p \times n$, then AB is $m \times n$ and

$$(AB)_{ij} = \sum_{k=1}^{p} a_{ik}b_{kj}.$$

Matrix multiplication satisfies

$$A(BC) = (AB)C, \qquad A(B + C) = AB + AC,$$

but it is generally not commutative: usually $AB \neq BA$.

Common matrix multiplication mistakes

The product AB is defined only when the number of columns of A equals the number of rows of B. Even when both AB and BA are defined, they may have different sizes or different entries. Thus one should never swap matrix order unless a theorem or computation justifies it.

The transpose of an $m \times n$ matrix A is the $n \times m$ matrix A^T obtained by interchanging rows and columns:

$$(A^T)_{ij} = a_{ji}.$$

3.2 Special Matrices

A triangular matrix has all entries above or below the main diagonal equal to zero. A lower triangular matrix has zeros above the main diagonal, and an upper triangular matrix has zeros below it. A diagonal matrix has all entries off the main diagonal equal to zero. A scalar matrix is a diagonal matrix whose diagonal entries are all equal. If the common diagonal entry is 1, the matrix is the identity matrix I.

3.3 Systems of Linear Algebraic Equations

A system of m linear equations in n variables has the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

The numbers $b_1, \ldots, b_m$ are constants. If all constants are zero, the system is homogeneous; otherwise it is inhomogeneous. A system is consistent if it has at least one solution and inconsistent if it has no solution.

The augmented matrix is

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right).$$

Elementary Operations

Equivalent systems are obtained by:

1. multiplying an equation by a nonzero constant;
2. interchanging two equations;
3. adding a nonzero multiple of one equation to another equation.

The corresponding elementary row operations for matrices are:

1. multiplying a row by a nonzero constant;
2. interchanging any two rows;
3. adding a nonzero multiple of one row to another row.

A row-echelon form satisfies: the first nonzero entry in a nonzero row is 1; in consecutive nonzero rows, the leading 1 in the lower row appears to the right of the leading 1 in the higher row; and rows of zeros are at the bottom. Gaussian elimination row-reduces to row-echelon form. Gauss–Jordan elimination row-reduces to reduced row-echelon form, where every column containing a leading 1 has zeros everywhere else.

3.4 Rank of a Matrix

The row vectors of an $m \times n$ matrix A are its rows; the column vectors are its columns. The rank of A, denoted $\text{rank}(A)$, is the maximum number of linearly independent row vectors in A. It can be found by row-reducing A to a row-echelon matrix B:

$$\text{rank}(A) = \text{the number of nonzero rows in } B.$$

A linear system $AX = B$ is consistent if and only if the rank of the coefficient matrix A equals the rank of the augmented matrix $(A|B)$. If the system is consistent, has n variables, and $\text{rank}(A) = r$, then the solution contains $n - r$ parameters.

Key Takeaways

- Matrix multiplication represents composition of linear actions and must be performed in order.
- Row operations preserve the solution set of a linear system.
- Rank compares the amount of independent information in the coefficient and augmented matrices.

Chapter 4

Determinants, Inverses, Eigenvalues, and Factorizations

Chapter Roadmap

Determinants measure invertibility, oriented area/volume scaling, and singularity. Inverses solve square systems, eigenvalues reveal preserved directions, and factorizations such as LU separate a difficult computation into simpler triangular systems.

4.1 Determinants

The determinant is a scalar value associated with a square matrix. Although it is defined by an algebraic formula, its most important meaning is geometric: it measures how a linear transformation changes area or volume.

For a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the columns of A are the vectors

$$\mathbf{u} = \begin{pmatrix} a \\ c \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} b \\ d \end{pmatrix}.$$

These two vectors span a parallelogram in the plane. The absolute value

$$|\det A|$$

is the area of this parallelogram. The sign of $\det A$ records orientation: a positive determinant preserves orientation, while a negative determinant reverses orientation.

Determinant of a 2×2 Matrix

For

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix},$$

the determinant is

$$\det A = a_{11}a_{22} - a_{12}a_{21}.$$

Geometric Meaning

For a 2×2 matrix A, the determinant tells how areas are scaled by the linear transformation $T(\mathbf{x}) = A\mathbf{x}$:

$$\text{area of transformed region} = |\det A| \cdot \text{area of original region}.$$

Thus:

- $|\det A| > 1$ means areas are expanded.
- $0 < |\det A| < 1$ means areas are contracted.
- $\det A = 0$ means area collapses to zero.
- $\det A < 0$ means orientation is reversed.

For a 3×3 matrix, the determinant has an analogous volume meaning. If the columns of A are vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$, then

$$|\det A|$$

is the volume of the parallelepiped spanned by $\mathbf{u}, \mathbf{v}, \mathbf{w}$. The sign again represents orientation.

Determinant and Invertibility

For a square matrix A,

$$\det A \neq 0$$

if and only if A is invertible. Equivalently, the linear transformation $T(\mathbf{x}) = A\mathbf{x}$ does not collapse space into a lower-dimensional object.
If

$$\det A = 0,$$

then A is singular, its columns are linearly dependent, and the associated linear transformation collapses area or volume to zero.

4.1.1 Cofactor Expansion

For a 3×3 matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

cofactor expansion along the first row gives

$$\det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13},$$

where

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Here M_{ij} is the determinant of the submatrix obtained by deleting row i and column j from A. It is called the minor corresponding to a_{ij}.

For an $n \times n$ matrix, cofactor expansion along row i is

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in},$$

and cofactor expansion along column j is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

For a 3×3 matrix, cofactor expansion reduces the determinant computation to several 2×2 determinant computations. For example, expansion along the first row means that each entry in the first row is multiplied by the determinant of the 2×2 matrix obtained by deleting that entry's row and column. The signs along the first row alternate as

$$+ \quad - \quad +.$$

Computing a 3×3 Determinant by Cofactor Expansion

Consider

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ 1 & 2 & -2 \end{pmatrix}.$$

We compute $\det A$ by expanding along the first row.

$$\det A = 2 \begin{vmatrix} 4 & 5 \\ 2 & -2 \end{vmatrix} - (-1) \begin{vmatrix} 0 & 5 \\ 1 & -2 \end{vmatrix} + 3 \begin{vmatrix} 0 & 4 \\ 1 & 2 \end{vmatrix}.$$

Each 2×2 determinant is computed by

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Hence,

$$\begin{vmatrix} 4 & 5 \\ 2 & -2 \end{vmatrix} = -18, \qquad \begin{vmatrix} 0 & 5 \\ 1 & -2 \end{vmatrix} = -5, \qquad \begin{vmatrix} 0 & 4 \\ 1 & 2 \end{vmatrix} = -4.$$

Therefore,

$$\det A = 2(-18) - (-1)(-5) + 3(-4) = -36 - 5 - 12 = -53.$$

$$\boxed{\det A = -53}$$

4.1.2 Basic Properties

For square matrices A and B of the same size:

$$\det(A^T) = \det A,$$

$$\det(AB) = \det A \det B.$$

If A is invertible, then

$$\det(A^{-1}) = \frac{1}{\det A}.$$

Row operations affect determinants as follows:

- Interchanging two rows changes the sign of the determinant.
- Multiplying one row by a scalar k multiplies the determinant by k.
- Adding a multiple of one row to another row does not change the determinant.

Why Determinant of A equals zero Matters

The condition $\det A = 0$ means that the column vectors of A fail to span the full space. In $\mathbb{R}^2$, the parallelogram formed by the columns has zero area, so the two column vectors lie on the same line. In $\mathbb{R}^3$, the parallelepiped formed by the columns has zero volume, so the three column vectors lie in a common plane or lower-dimensional subspace.

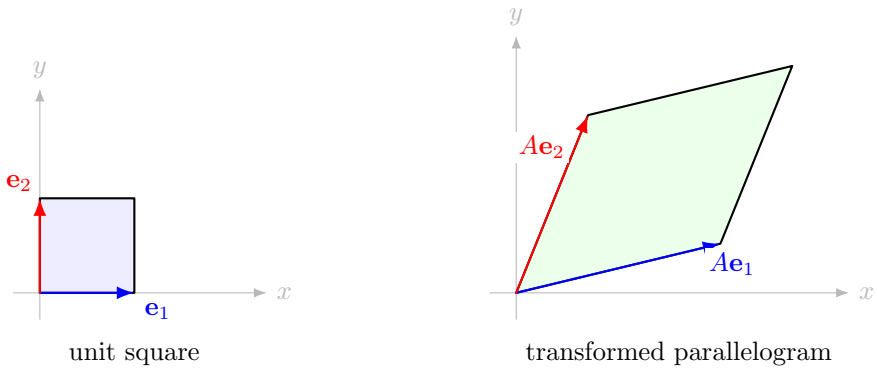

Figure 4.1: A 2×2 matrix A transforms the unit square into the parallelogram spanned by its column vectors $A\mathbf{e}_1$ and $A\mathbf{e}_2$. Since the unit square has area 1, the area of the transformed parallelogram is $|\det A|$. Thus $|\det A|$ measures the area scaling factor of A.

4.2 Inverse Matrices

Definition

If A is an $n \times n$ matrix and there exists an $n \times n$ matrix B such that

$$AB = BA = I,$$

then A is nonsingular or invertible, and B is the inverse of A, denoted A^{-1}.

Theorem / Property

For nonsingular matrices A and B,

$$(A^{-1})^{-1} = A, \qquad (AB)^{-1} = B^{-1}A^{-1}, \qquad (A^T)^{-1} = (A^{-1})^T.$$

The adjoint matrix is the transpose of the cofactor matrix:

$$\operatorname{adj} A = (C_{ij})^T.$$

The inverse can be expressed by

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A, \qquad \det A \neq 0.$$

If a square matrix A can be transformed into I by elementary row operations, then A is nonsingular, and the same operations transform I into A^{-1}. Equivalently, row-reduce $[A|I]$ to $[I|A^{-1}]$.

A linear system $AX = B$ with nonsingular A has solution

$$X = A^{-1}B.$$

A homogeneous system $AX = 0$ has only the trivial solution if and only if A is nonsingular.

4.3 Cramer's Rule

Cramer's Rule is a determinant-based method for solving a square linear system

$$A\mathbf{x} = \mathbf{b},$$

where A is an $n \times n$ coefficient matrix, $\mathbf{x}$ is the unknown vector, and $\mathbf{b}$ is the constant vector. The method expresses each unknown x_k as a ratio of determinants.

Cramer's Rule

Consider a system of n linear equations in n variables:

$$A\mathbf{x} = \mathbf{b}.$$

Suppose that A is nonsingular, equivalently

$$\det A \neq 0.$$

For each $k = 1, 2, \ldots, n$, let A_k be the matrix obtained from A by replacing the k-th column of A with the constant vector $\mathbf{b}$. Then the unique solution is given by

$$x_k = \frac{\det A_k}{\det A}, \qquad k = 1, 2, \ldots, n.$$

When Cramer's Rule Is Useful

Cramer's Rule is most useful when:

- the system has only two or three variables;
- one wants an exact symbolic formula for a particular variable;
- the determinant of the coefficient matrix is easy to compute;
- the goal is theoretical understanding rather than efficient computation.

For large systems, Cramer's Rule is usually inefficient because it requires computing many determinants. In practice, Gaussian elimination, LU-factorization, or matrix inverse methods are usually preferred.

Solving a 2×2 System by Cramer's Rule

Solve

$$\begin{cases} 2x + y = 5, \\ x - 3y = -4. \end{cases}$$

The coefficient matrix and constant vector are

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}.$$

First compute

$$\det A = \begin{vmatrix} 2 & 1 \\ 1 & -3 \end{vmatrix} = 2(-3) - 1(1) = -7.$$

Since $\det A \neq 0$, the system has a unique solution and Cramer's Rule applies.

To find x, replace the first column of A by $\mathbf{b}$:

$$A_1 = \begin{pmatrix} 5 & 1 \\ -4 & -3 \end{pmatrix}.$$

Thus

$$x = \frac{\det A_1}{\det A} = \frac{\begin{vmatrix} 5 & 1 \\ -4 & -3 \end{vmatrix}}{-7} = \frac{5(-3) - 1(-4)}{-7} = \frac{-11}{-7} = \frac{11}{7}.$$

To find y, replace the second column of A by $\mathbf{b}$:

$$A_2 = \begin{pmatrix} 2 & 5 \\ 1 & -4 \end{pmatrix}.$$

Thus

$$y = \frac{\det A_2}{\det A} = \frac{\begin{vmatrix} 2 & 5 \\ 1 & -4 \end{vmatrix}}{-7} = \frac{2(-4) - 5(1)}{-7} = \frac{-13}{-7} = \frac{13}{7}.$$

Therefore,

$$\boxed{x = \frac{11}{7}, \qquad y = \frac{13}{7}.}$$

4.4 Eigenvalue Problems

A matrix can be understood not only as an array of numbers, but also as a linear transformation. When a matrix A acts on a vector $\mathbf{v}$, it usually changes both the length and the direction of $\mathbf{v}$. However, some special nonzero vectors keep their direction after the transformation. These special vectors are called *eigenvectors*.

Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix. A nonzero vector $\mathbf{v} \in \mathbb{R}^n$ is called an *eigenvector* of A if there exists a scalar λ such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The scalar λ is called the *eigenvalue* corresponding to the eigenvector $\mathbf{v}$.

The equation

$$A\mathbf{v} = \lambda\mathbf{v}$$

means that applying A to $\mathbf{v}$ gives a vector parallel to $\mathbf{v}$. Therefore, an eigenvector is a direction that is preserved by the linear transformation A. The eigenvalue λ tells how the vector is scaled:

λ	Geometric meaning
$\lambda > 1$	the vector is stretched
$0 < \lambda < 1$	the vector is shortened
$\lambda < 0$	the vector is reversed in direction and scaled
$\lambda = 0$	the vector is sent to the zero vector

Important distinction

The zero vector is never called an eigenvector. This is because

$$A\mathbf{0} = \lambda\mathbf{0}$$

holds for every scalar λ, so it would not identify any special direction. Eigenvectors must be nonzero.

To find possible eigenvalues, rewrite the eigenvalue equation as

$$A\mathbf{v} = \lambda\mathbf{v} \quad \Longleftrightarrow \quad A\mathbf{v} - \lambda I\mathbf{v} = \mathbf{0}.$$

Since $\lambda I\mathbf{v} = \lambda\mathbf{v}$, this becomes

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

This is a homogeneous linear system. We want a nonzero solution $\mathbf{v} \neq \mathbf{0}$, so the coefficient matrix $A - \lambda I$ must be singular. Hence

$$\det(A - \lambda I) = 0.$$

Characteristic Equation

The equation

$$\det(A - \lambda I) = 0$$

is called the *characteristic equation* of A. Its solutions are the eigenvalues of A.

Once an eigenvalue λ is found, the corresponding eigenvectors are obtained by solving

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

The set of all solutions to this equation is called the *eigenspace* corresponding to λ:

$$E_\lambda = \{\mathbf{v} \in \mathbb{R}^n : (A - \lambda I)\mathbf{v} = \mathbf{0}\}.$$

The nonzero vectors in E_λ are the eigenvectors corresponding to λ.

How to Find Eigenvalues and Eigenvectors

To solve an eigenvalue problem for an $n \times n$ matrix A:

1. Form the matrix $A - \lambda I$.
2. Compute the characteristic equation
$$\det(A - \lambda I) = 0.$$
3. Solve this equation to find the eigenvalues λ.
4. For each eigenvalue λ, solve
$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$
using row reduction.
5. The nonzero solution vectors are the eigenvectors corresponding to λ.

Important Facts About Eigenvalues

- $\lambda = 0$ is an eigenvalue of A if and only if A is singular.
- If λ is an eigenvalue of an invertible matrix A with eigenvector $\mathbf{v}$, then λ^{-1} is an eigenvalue of A^{-1} with the same eigenvector.
- If A is triangular, then its eigenvalues are the diagonal entries of A.

Eigenvectors have an important geometric interpretation: under the linear transformation determined by A, an eigenvector changes only by a scalar factor. Its direction is preserved, although its length may change and its orientation may reverse if the eigenvalue is negative.

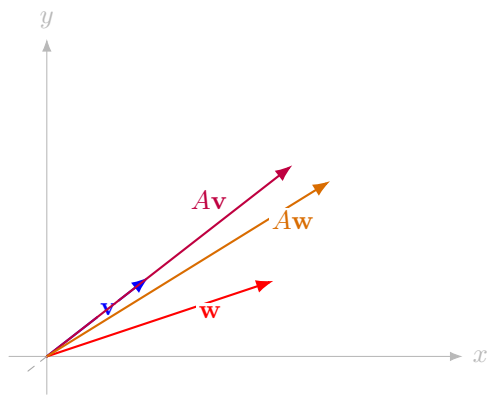

Figure 4.2: An eigenvector is a direction preserved by a linear transformation. The vectors $\mathbf{v}$ and $A\mathbf{v}$ lie on the same line, so $\mathbf{v}$ is an eigenvector and $A\mathbf{v} = \lambda\mathbf{v}$ for some scalar λ. By contrast, $\mathbf{w}$ and $A\mathbf{w}$ do not have the same direction, so $\mathbf{w}$ is not an eigenvector.

4.5 Powers of Matrices and the Cayley–Hamilton Theorem

Computing high powers of a matrix directly can be expensive. The Cayley–Hamilton theorem gives a systematic way to reduce high powers of a square matrix to lower powers. The key idea is that a matrix satisfies the same polynomial equation as its eigenvalues.

Cayley–Hamilton Theorem

Let A be an $n \times n$ matrix. Suppose the characteristic equation of A is

$$p(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0 = 0.$$

Then A satisfies its own characteristic equation:

$$p(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I = 0.$$

The theorem is useful because it lets us rewrite high powers of A using only lower powers. In particular, for an $n \times n$ matrix, every sufficiently high power of A can be expressed as a linear combination of

$$I, A, A^2, \ldots, A^{n-1}.$$

For a 2×2 matrix, this means every power A^m with $m \geq 2$ can be rewritten using only I and A. This idea is useful in matrix recurrences, linear differential equations, repeated linear transformations, and matrix exponentials.

For example, suppose the characteristic equation of a 2×2 matrix A is

$$\lambda^2 - 5\lambda + 6 = 0.$$

By the Cayley–Hamilton theorem,

$$A^2 - 5A + 6I = 0,$$

so

$$A^2 = 5A - 6I.$$

Multiplying both sides by A, we get

$$A^3 = 5A^2 - 6A.$$

Substituting $A^2 = 5A - 6I$ gives

$$A^3 = 5(5A - 6I) - 6A = 19A - 30I.$$

Thus the higher power A^3 has been rewritten using only A and I.

4.6 Diagonalizability

Diagonalization is a way of rewriting a matrix in a simpler coordinate system. The central idea is that if a matrix has enough eigenvectors, then the matrix acts like a diagonal matrix after a suitable change of basis.

An $n \times n$ matrix A is called *diagonalizable* if there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

Equivalently,

$$P^{-1}AP = D.$$

The columns of P are eigenvectors of A, and the corresponding diagonal entries of D are the eigenvalues. More precisely, if

$$P = \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{pmatrix},$$

where $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent eigenvectors of A, then

$$D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix},$$

where each λ_i is the eigenvalue corresponding to $\mathbf{v}_i$.

The fundamental criterion is simple: an $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors. In particular, if an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable. The converse is not necessarily true: a matrix can still be diagonalizable even when some eigenvalues are repeated.

To diagonalize a matrix A, first solve

$$\det(A - \lambda I) = 0$$

to find the eigenvalues. Then, for each eigenvalue λ, solve

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

to find the corresponding eigenspace. If the collected eigenvectors give n linearly independent vectors, place them as the columns of P. The diagonal matrix D is formed by placing the matching eigenvalues on the diagonal in the same order as the eigenvectors in P. Then

$$A = PDP^{-1}.$$

Diagonalization is especially useful for matrix powers. If

$$A = PDP^{-1},$$

then

$$A^m = PD^mP^{-1}.$$

Since D is diagonal, D^m is obtained simply by raising each diagonal entry to the m-th power. This is one of the main reasons diagonalization is important in applications involving repeated transformations or recurrence relations.

4.7 Orthogonal Matrices and Orthogonal Diagonalization

Orthogonal matrices are matrices whose columns form an orthonormal basis. A square matrix Q is called *orthogonal* if

$$Q^{-1} = Q^T.$$

Equivalently,

$$Q^TQ = I.$$

This condition means that the columns of Q are mutually orthogonal unit vectors.

Orthogonal matrices are important because they preserve lengths and angles. If Q is orthogonal, then for any vectors $\mathbf{x}$ and $\mathbf{y}$,

$$\|Q\mathbf{x}\| = \|\mathbf{x}\|, \qquad (Q\mathbf{x}) \cdot (Q\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}.$$

Thus orthogonal transformations behave like rotations, reflections, or combinations of these: they may change the position or orientation of vectors, but they do not distort lengths or angles.

A matrix A is called *symmetric* if

$$A = A^T.$$

Symmetric matrices have especially nice eigenvalue behavior. Their eigenvalues are real, and eigenvectors corresponding to distinct eigenvalues are orthogonal. This leads to one of the most important results about real symmetric matrices.

Orthogonal Diagonalization Theorem

A real $n \times n$ matrix A is orthogonally diagonalizable if and only if A is symmetric. Equivalently, A is symmetric if and only if there exists an orthogonal matrix Q and a diagonal matrix D such that

$$Q^TAQ = D.$$

In this case,

$$A = QDQ^T.$$

To orthogonally diagonalize a real symmetric matrix A, first find its eigenvalues and corresponding eigenvectors. Eigenvectors coming from distinct eigenvalues are already orthogonal.

If an eigenvalue is repeated, one chooses an orthogonal basis for that eigenspace, using Gram–Schmidt if necessary. After all eigenvectors are chosen, normalize them to have length 1, place them as the columns of an orthogonal matrix Q, and place the corresponding eigenvalues on the diagonal of D. Then

$$Q^T AQ = D.$$

Orthogonal diagonalization is often better than ordinary diagonalization because

$$Q^{-1} = Q^T.$$

Thus

$$A = QDQ^T$$

is usually simpler and more stable than

$$A = PDP^{-1}.$$

In numerical computation, using a transpose is generally easier and more reliable than computing a general inverse.

4.8 LU-Factorization

LU-factorization is a method for rewriting a matrix as a product of a lower triangular matrix and an upper triangular matrix. If a square matrix A can be written as

$$A = LU,$$

where L is lower triangular and U is upper triangular, then this is called an *LU-factorization* or *LU-decomposition* of A.

The main purpose of LU-factorization is to solve linear systems efficiently. Instead of solving

$$A\mathbf{x} = \mathbf{b}$$

directly, we write $A = LU$, so the system becomes

$$LU\mathbf{x} = \mathbf{b}.$$

Let

$$U\mathbf{x} = \mathbf{y}.$$

Then the original system is split into two triangular systems:

$$L\mathbf{y} = \mathbf{b}, \qquad U\mathbf{x} = \mathbf{y}.$$

The first system is solved by forward substitution, and the second is solved by back substitution. Triangular systems are much easier to solve than general linear systems.

LU-factorization is especially useful when solving many systems with the same coefficient matrix but different right-hand sides:

$$A\mathbf{x}_1 = \mathbf{b}_1, \quad A\mathbf{x}_2 = \mathbf{b}_2, \quad A\mathbf{x}_3 = \mathbf{b}_3, \quad \ldots$$

Once the factorization $A = LU$ has been computed, each new system only requires forward and back substitution.

A common way to construct $A = LU$ is to perform Gaussian elimination on A until an upper triangular matrix U is obtained. The multipliers used to eliminate entries below the diagonal

are stored in the lower triangular matrix L. In the Doolittle convention, the diagonal entries of L are all 1:

$$L = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \ell_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{n1} & \ell_{n2} & \cdots & 1 \end{pmatrix}.$$

For example, let

$$A = \begin{pmatrix} 2 & 1 \\ 4 & 5 \end{pmatrix}.$$

To eliminate the entry below the first pivot, use

$$R_2 \leftarrow R_2 - 2R_1.$$

This gives

$$U = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

The multiplier used was 2, so

$$L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Indeed,

$$LU = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 5 \end{pmatrix} = A.$$

Therefore,

$$A = LU.$$

Key Takeaways

- Determinants identify singularity and support cofactor computation.
- Inverse matrices solve square systems when $\det A \neq 0$.
- Eigenvectors are directions preserved by a matrix; eigenvalues are the corresponding scale factors.

Chapter 5

Laplace Transforms

Chapter Roadmap

The Laplace transform converts differential equations into algebraic equations. The central skill is to transform an initial-value problem, solve for the transformed unknown, and then apply inverse transforms to recover the time-domain solution.

5.1 Definition and Linearity

Definition

If $f(t)$ is defined for $t \geq 0$, its Laplace transform is

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt,$$

provided the integral converges. The result $F(s)$ is a function of s.

The Laplace transform is linear:

$$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\} = aF(s) + bG(s).$$

5.2 Basic Transform Table

Function	Laplace transform
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}, \quad n = 1, 2, 3, \ldots$
e^{at}	$\frac{1}{s-a}$
$\sin at$	$\frac{a}{s^2+a^2}$
$\cos at$	$\frac{s}{s^2+a^2}$
$\sinh at$	$\frac{a}{s^2-a^2}$
$\cosh at$	$\frac{s}{s^2-a^2}$

The inverse Laplace transform satisfies

$$f(t) = \mathcal{L}^{-1}\{F(s)\},$$

and is also linear:

$$\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}\{F(s)\} + b\mathcal{L}^{-1}\{G(s)\}.$$

5.3 Transforms of Derivatives

If $F(s) = \mathcal{L}\{f(t)\}$, then

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0),$$
$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0),$$
$$\mathcal{L}\{f'''(t)\} = s^3F(s) - s^2f(0) - sf'(0) - f''(0),$$

and in general

$$\mathcal{L}\{f^{(n)}(t)\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0).$$

5.4 Solving Initial-Value Problems

The Laplace-transform method for a linear ordinary differential equation is:

1. Start with an unknown $y(t)$ satisfying a differential equation and initial conditions.
2. Apply the Laplace transform $\mathcal{L}$ to convert the differential equation into an algebraic equation in $Y(s)$.
3. Solve the algebraic equation for $Y(s)$.
4. Apply the inverse transform $\mathcal{L}^{-1}$ to recover $y(t)$.

5.5 Shift on the s-Axis

For $F(s) = \mathcal{L}\{f(t)\}$,

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a).$$

This is the first shifting theorem.

5.6 Unit Step Function

The unit step function is

$$U(t-a) = \begin{cases} 0, & 0 \le t < a, \\ 1, & t \ge a. \end{cases}$$

It is used to represent functions that turn on at time a and is central to second-shifting formulas and piecewise forcing functions.

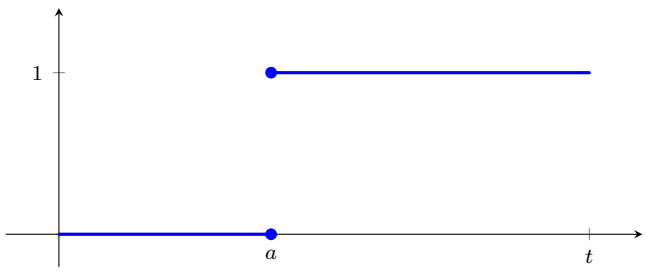

Figure 5.1: The unit step function $U(t-a)$ switches from 0 to 1 at $t = a$.

Key Takeaways

- Laplace transforms turn derivatives into polynomials in s plus initial-value terms.
- Initial-value problems become algebraic equations for $Y(s)$.
- Shift theorems handle exponential factors and delayed step functions.

Chapter 6

Vector Functions, Motion, and Partial Derivatives

Chapter Roadmap

Vector functions describe curves and motion, while partial derivatives describe how multivariable functions change in one variable at a time. The chapter connects geometry, motion, chain rules, and local rates of change.

6.1 Vector Functions

A vector $\mathbf{r}$ is a vector function when its components depend on a parameter t:

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}.$$

As t varies, the terminal point of $\mathbf{r}(t)$ moves; therefore a curve can be described by a vector function.

The derivative is

$$\mathbf{r}'(t) = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \langle f'(t), g'(t), h'(t) \rangle.$$

If $\mathbf{r}'(t) \neq \mathbf{0}$ at point P, then $\mathbf{r}'(t)$ is tangent to the curve at P. The tangent line is the line through P parallel to $\mathbf{r}'(t)$.

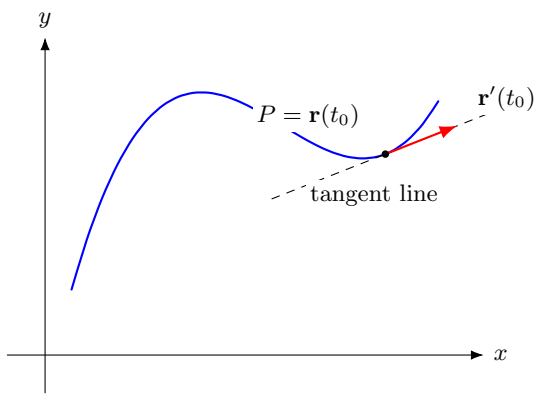

Figure 6.1: A vector-valued function traces out a curve, and the derivative vector at a point gives the tangent direction.

If the components are integrable,

$$\int \mathbf{r}(t)\,dt = \left(\int f(t)\,dt\right)\mathbf{i} + \left(\int g(t)\,dt\right)\mathbf{j} + \left(\int h(t)\,dt\right)\mathbf{k}.$$

Rules of Differentiation

Let $\mathbf{r}_1, \mathbf{r}_2$ be differentiable vector functions and let $u(t)$ be a differentiable scalar function. Then

$$\begin{aligned}
\frac{d}{dt}\left[\mathbf{r}_1(t) + \mathbf{r}_2(t)\right] &= \mathbf{r}_1'(t) + \mathbf{r}_2'(t),\\
\frac{d}{dt}\left[u(t)\mathbf{r}_1(t)\right] &= u(t)\mathbf{r}_1'(t) + u'(t)\mathbf{r}_1(t),\\
\frac{d}{dt}\left[\mathbf{r}_1(t)\cdot\mathbf{r}_2(t)\right] &= \mathbf{r}_1(t)\cdot\mathbf{r}_2'(t) + \mathbf{r}_1'(t)\cdot\mathbf{r}_2(t),\\
\frac{d}{dt}\left[\mathbf{r}_1(t)\times\mathbf{r}_2(t)\right] &= \mathbf{r}_1(t)\times\mathbf{r}_2'(t) + \mathbf{r}_1'(t)\times\mathbf{r}_2(t).
\end{aligned}$$

The chain rule states that if $s = u(t)$, then

$$\frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds}\frac{ds}{dt} = \mathbf{r}'(s)u'(t).$$

6.2 Length of a Curve

If

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}, \qquad a \le t \le b,$$

then the curve length is

$$s = \int_a^b \|\mathbf{r}'(t)\|\,dt = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2}\,dt.$$

6.3 Motion on a Curve

For a body moving along a curve,

$$\text{position: } \mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k},$$

$$\text{velocity: } \mathbf{v}(t) = \mathbf{r}'(t), \qquad \text{acceleration: } \mathbf{a}(t) = \mathbf{r}''(t).$$

The speed is

$$\|\mathbf{v}(t)\| = \left\|\frac{d\mathbf{r}}{dt}\right\|.$$

For circular motion in a plane,

$$\mathbf{r}(t) = r_0\cos(\omega t)\mathbf{i} + r_0\sin(\omega t)\mathbf{j},$$

and

$$\mathbf{a}(t) = \mathbf{r}''(t) = -\omega^2\mathbf{r}(t).$$

For curvilinear projectile motion under gravity,

$$\mathbf{a}(t) = -g\mathbf{j},$$

with initial velocity

$$\mathbf{v}_0 = v_0 \cos\theta\,\mathbf{i} + v_0 \sin\theta\,\mathbf{j}$$

and initial position

$$\mathbf{r}_0 = s_0\mathbf{j}.$$

Integrating acceleration gives

$$\mathbf{v}(t) = v_0 \cos\theta\,\mathbf{i} + (-gt + v_0 \sin\theta)\mathbf{j},$$

and integrating velocity gives

$$\mathbf{r}(t) = (v_0 \cos\theta)t\mathbf{i} + \left(-\frac{1}{2}gt^2 + v_0 \sin\theta\, t + s_0\right)\mathbf{j}.$$

6.4 Partial Derivatives

If $z = f(x, y)$, the partial derivative with respect to x is

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x},$$

and the partial derivative with respect to y is

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}.$$

Common notation includes

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = f_x = D_x f, \qquad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = f_y = D_y f.$$

The second-order partial derivatives are

$$f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2},$$

$$f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x}, \quad f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y}.$$

6.5 Chain Rule for Several Variables

If $y = f(u)$ and $u = g(x)$, then the ordinary chain rule is

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

For $z = f(u, v)$ where $u = u(x, y)$ and $v = v(x, y)$,

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x},$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial y}.$$

Key Takeaways

- Vector functions describe curves; derivatives give tangent directions, velocity, and acceleration.
- Partial derivatives measure change in one variable while holding the other variables fixed.
- The gradient packages directional derivatives into one vector.

Chapter 7

Line Integrals, Conservative Fields, Double Integrals, and Green's Theorem

Chapter Roadmap

Line integrals accumulate a vector field along a curve, while double integrals accumulate scalar quantities over regions. Conservative fields introduce path independence, and Green's theorem connects circulation around a boundary with a double integral over the enclosed region.

7.1 Line Integrals

The work done by a force field $\mathbf{F}$ as an object moves from A to B along a curve C is approximated by

$$W \approx \sum_{k=1}^{n} \mathbf{F}_k \cdot \Delta \mathbf{r}_k.$$

Taking the limit gives the line integral

$$W = \int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

If

$$\mathbf{F} = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j},$$

then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P(x, y)\, dx + Q(x, y)\, dy.$$

If

$$\mathbf{F} = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k},$$

then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P\, dx + Q\, dy + R\, dz.$$

A line integral of a vector field around a closed path is called the circulation of the field around the path.

7.2 Conservative Fields

If the circulation of $\mathbf{F}$ around a closed curve is nonzero, the field is nonconservative. A conservative field has path-independent line integrals: if C_1 and C_2 connect the same initial and terminal points, then

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}.$$

For a conservative field, there exists a potential function ϕ such that

$$\mathbf{F} = \nabla \phi.$$

Then

$$\int_A^B \mathbf{F} \cdot d\mathbf{r} = \phi(B) - \phi(A).$$

For a planar field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ on a simply connected region with continuous first partial derivatives, a standard test is

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \Longleftrightarrow \quad \mathbf{F} \text{ is conservative.}$$

Important domain condition

The equality $P_y = Q_x$ alone is not enough on every possible domain. The usual "easy test" requires the field to be defined on a region without holes, such as a disk or rectangle. On regions with holes, one must be more careful and may need to evaluate circulation directly.

7.3 Double Integrals

For a Type I region

$$R = \{(x, y) : a \le x \le b,\ g_1(x) \le y \le g_2(x)\},$$

$$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx.$$

For a Type II region

$$R = \{(x, y) : c \le y \le d,\ h_1(y) \le x \le h_2(y)\},$$

$$\iint_R f(x, y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\, dy.$$

7.4 Polar Coordinates

In polar coordinates,

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad dA = r\, dr\, d\theta.$$

If

$$R = \{(r, \theta) : \alpha \le \theta \le \beta,\ g_1(\theta) \le r \le g_2(\theta)\},$$

then

$$\iint_R f(r, \theta)\, dA = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} f(r, \theta) r\, dr\, d\theta.$$

If

$$R = \{(r, \theta) : a \leq r \leq b,\ h_1(r) \leq \theta \leq h_2(r)\},$$

then

$$\iint_R f(r, \theta)\, dA = \int_a^b \int_{h_1(r)}^{h_2(r)} f(r, \theta) r\, d\theta\, dr.$$

7.5 Green's Theorem

Let

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}.$$

If $P, Q, \partial P/\partial y$, and $\partial Q/\partial x$ are continuous on a region R and its positively oriented simple closed boundary C, then Green's theorem states

$$\oint_C P\, dx + Q\, dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Positive orientation means counterclockwise orientation of the boundary; negative orientation means clockwise orientation.

Key Takeaways

- Line integrals compute work and circulation.
- Conservative fields are path independent when a potential function exists.
- Green's theorem converts boundary circulation into a double integral over the enclosed region.

Chapter 8

Complex Numbers and Complex Functions

Chapter Roadmap

Complex numbers combine algebra, geometry, and calculus in the plane. This chapter covers rectangular and polar forms, roots, complex functions, differentiability, the Cauchy–Riemann equations, and the fundamental contour integral theorems.

8.1 Complex Numbers

A complex number has the form

$$z = x + \mathrm{i}y,$$

where $x, y \in \mathbb{R}$ and $\mathrm{i}^2 = -1$. The real part is $\Re z = x$, and the imaginary part is $\Im z = y$.

For $z_1 = x_1 + \mathrm{i}y_1$ and $z_2 = x_2 + \mathrm{i}y_2$,

$$z_1 + z_2 = (x_1 + x_2) + \mathrm{i}(y_1 + y_2),$$

$$z_1 - z_2 = (x_1 - x_2) + \mathrm{i}(y_1 - y_2),$$

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + \mathrm{i}(x_1 y_2 + y_1 x_2),$$

$$\frac{z_1}{z_2} = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + \mathrm{i}\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}, \qquad z_2 \neq 0.$$

The conjugate of $z = x + \mathrm{i}y$ is

$$z^* = \overline{z} = x - \mathrm{i}y.$$

The modulus is

$$|z| = \sqrt{x^2 + y^2} = \sqrt{z\overline{z}}.$$

8.2 Powers and Roots

A complex number can be expressed in polar form as

$$z = r(\cos\theta + \mathrm{i}\sin\theta),$$

where $r = |z|$ and $\theta = \arg z$. Integer powers are given by De Moivre's formula:

$$z^n = r^n(\cos n\theta + \mathrm{i}\sin n\theta).$$

Multiplication and division in polar form are

$$z_1 z_2 = r_1 r_2 \big[\cos(\theta_1 + \theta_2) + \mathrm{i}\sin(\theta_1 + \theta_2)\big],$$
$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\big[\cos(\theta_1 - \theta_2) + \mathrm{i}\sin(\theta_1 - \theta_2)\big].$$

A number w is an nth root of nonzero z if $w^n = z$. If $z = r(\cos\theta + \mathrm{i}\sin\theta)$, then the n distinct roots are

$$w_k = r^{1/n}\left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + \mathrm{i}\sin\left(\frac{\theta + 2\pi k}{n}\right)\right], \quad k = 0, 1, \ldots, n-1.$$

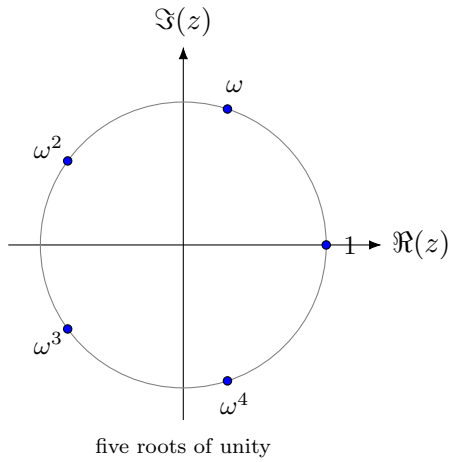

Figure 8.1: Roots of unity in the complex plane.

8.3 Functions of a Complex Variable

A function of a complex variable $z = x + \mathrm{i}y$ can be written as

$$w = f(z) = u(x, y) + \mathrm{i}v(x, y),$$

where u is the real part and v is the imaginary part. A complex function can be interpreted as a mapping or transformation from the z-plane to the w-plane.

8.4 Limits, Continuity, and Derivatives

For a real function, $\lim_{x\to x_0} f(x) = L$ means $f(x)$ approaches L as x approaches x_0 from the left or right on the real line. For a complex function,

$$\lim_{z\to z_0} f(z) = L$$

means $f(z)$ approaches L as z approaches z_0 from any direction in the complex plane.

The function f is continuous at z_0 if

$$\lim_{z\to z_0} f(z) = f(z_0).$$

The derivative of f at z_0 is

$$f'(z_0) = \lim_{\Delta z\to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z},$$

provided the limit exists and is independent of the direction of approach.

A complex function is analytic at z_0 if it is differentiable at z_0 and at every point in some neighborhood of z_0.

8.5 Cauchy–Riemann Equations

For

$$f(z) = u(x, y) + \mathrm{i}v(x, y),$$

the Cauchy–Riemann equations are

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

If u and v have continuous first-order partial derivatives in a neighborhood of z, and if they satisfy the Cauchy–Riemann equations throughout that neighborhood, then f is analytic there. At a single point, the Cauchy–Riemann equations are necessary for differentiability, but by themselves they are not always sufficient unless suitable continuity assumptions are included.

8.6 Integration in the Complex Plane

For a complex function $f(z) = u(x, y) + \mathrm{i}v(x, y)$ and a contour C parametrized by $z = z(t)$ for $a \le t \le b$, the contour integral is

$$\int_C f(z)\,dz = \int_a^b f(z(t))z'(t)\,dt.$$

Cauchy–Goursat Theorem

If f is analytic at all points within and on a simple closed contour C, then

$$\int_C f(z)\,dz = 0.$$

Cauchy's Integral Formula

If f is analytic at all points within and on a simple closed contour C, and z_0 is any point interior to C, then

$$f(z_0) = \frac{1}{2\pi\mathrm{i}} \int_C \frac{f(z)}{z - z_0}\,dz,$$

equivalently,

$$\int_C \frac{f(z)}{z - z_0}\,dz = 2\pi\mathrm{i}f(z_0).$$

Key Takeaways

- Polar form turns multiplication into angle addition and powers into angle multiplication.
- Complex differentiability is stronger than real differentiability because limits must agree from every direction.
- Cauchy's theorems turn analyticity into powerful integral identities.

Chapter 9

Additional Operational and Multivariable Topics from the Slides

9.1 Additional Determinant Properties

For a square matrix A, the determinant satisfies the following basic properties:

1. $\det(A^T) = \det(A)$.
2. If any two rows or any two columns of A are the same, then $\det(A) = 0$.
3. If all entries in a row or column of A are zero, then $\det(A) = 0$.
4. If B is obtained by interchanging any two rows or columns of A, then $\det(B) = -\det(A)$.
5. Cofactor expansion can be performed along any row or column; choosing a row or column with many zeros minimizes computation.

9.2 Additional Laplace Transform Operations

The second shifting theorem gives

$$\mathcal{L}\{f(t-a)U(t-a)\} = e^{-as}F(s), \qquad F(s) = \mathcal{L}\{f(t)\}.$$

More generally,

$$\mathcal{L}\{g(t)U(t-a)\} = e^{-as}\mathcal{L}\{g(t+a)\}.$$

Multiplication by powers of t is converted into differentiation in the s-domain:

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}F(s), \qquad n = 1, 2, 3, \ldots.$$

The transform of an integral is

$$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{F(s)}{s},$$

so

$$\mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(\tau)\,d\tau.$$

9.3 Directional Derivatives and Gradients

For $z = f(x, y)$ and a unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$, the directional derivative is

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)u_1 + f_y(x, y)u_2.$$

The gradient is

$$\nabla f(x, y) = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j},$$

and therefore

$$D_{\mathbf{u}}f(x, y) = \nabla f(x, y) \cdot \mathbf{u}.$$

For $w = f(x, y, z)$ and a unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$,

$$D_{\mathbf{u}}f(x, y, z) = f_x u_1 + f_y u_2 + f_z u_3 = \nabla f(x, y, z) \cdot \mathbf{u}.$$

The gradient controls the maximum directional change:

$$-\|\nabla f\| \leq D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} \leq \|\nabla f\|.$$

Thus ∇f points in the direction of greatest increase, and $-\nabla f$ points in the direction of greatest decrease.

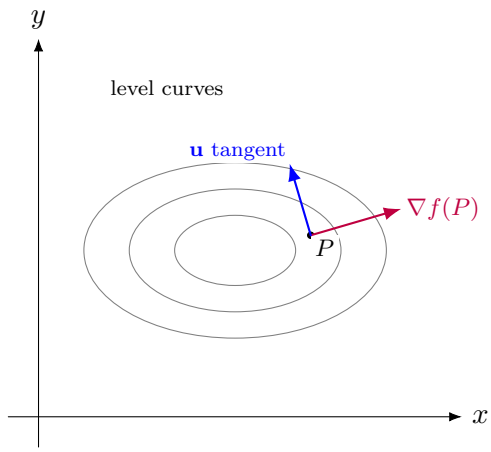

Figure 9.1: The gradient is perpendicular to a level curve and points in the direction of greatest increase.

9.4 Level Curves, Level Surfaces, Tangent Planes, and Normal Lines

Level curves are cross-sections of the graph $z = f(x, y)$ parallel to the xy-plane. Level surfaces are surfaces defined by

$$f(x, y, z) = c.$$

The gradient of f is normal to a level curve or level surface. If $P(x_0, y_0, z_0)$ is on the surface $f(x, y, z) = c$, then the tangent plane at P is

$$\nabla f(x_0, y_0, z_0) \cdot (\mathbf{r} - \mathbf{r}_0) = 0.$$

The normal line at P is the line through P parallel to

$$\nabla f(x_0, y_0, z_0).$$

Formula Summary

Vectors

$$\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i = \|\mathbf{a}\|\|\mathbf{b}\| \cos\theta$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\| \sin\theta$$

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$$

Lines and Planes

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{a}$$

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_1) = 0$$

Matrices

$$(AB)_{ij} = \sum_k a_{ik} b_{kj}$$

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$$

$$\det(A - \lambda I) = 0$$

$$AK = \lambda K$$

Laplace

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

$$\mathcal{L}\{f'\} = sF(s) - f(0)$$

$$\mathcal{L}\{e^{at} f(t)\} = F(s - a)$$

Vector Functions

$$\mathbf{v} = \mathbf{r}', \qquad \mathbf{a} = \mathbf{r}''$$

$$s = \int_a^b \|\mathbf{r}'(t)\| dt$$

Partial Derivatives

$$f_x = \frac{\partial f}{\partial x}, \qquad f_y = \frac{\partial f}{\partial y}$$

$$\frac{\partial z}{\partial x} = z_u u_x + z_v v_x$$

Line and Double Integrals

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy + R dz$$

$$\oint_C P dx + Q dy = \iint_R (Q_x - P_y) dA$$

$$dA = r dr d\theta$$

Complex Variables

$$z = x + \mathrm{i}y = r(\cos\theta + \mathrm{i}\sin\theta)$$

$$z^n = r^n(\cos n\theta + \mathrm{i}\sin n\theta)$$

$$u_x = v_y,$$
$$u_y = -v_x.$$

$$f(z_0) = \frac{1}{2\pi\mathrm{i}} \int_C \frac{f(z)}{z - z_0} dz$$